

# RESEARCH REPORT - COMPARATIVE ANALYSIS OF AI ALGORITHMS USED IN AGRICULTURE

Trinetra Drones and Robotics Pvt. Ltd.

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Submitted By Anshika Kodukula  
SRM UNIVERSITY-AP  
AP24110020029

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# Comparative Analysis of AI Algorithms Used in Agriculture

## Introduction

Artificial Intelligence (AI) has become a key technology in modern agriculture. AI algorithms help farmers monitor crop health, detect diseases, predict yields, optimize irrigation, and improve decision-making. Different machine learning and deep learning algorithms offer unique advantages depending on the agricultural application.

This report compares commonly used AI algorithms for agricultural applications, particularly crop disease detection using drone imagery.

| COMPARATIVE ANALYSIS OF AI ALGORITHMS USED IN AGRICULTURE |                     |                                                    |           |                  |                                                 |                                                       |
|-----------------------------------------------------------|---------------------|----------------------------------------------------|-----------|------------------|-------------------------------------------------|-------------------------------------------------------|
| Algorithm                                                 | Type                | Agricultural Application                           | Accuracy  | Processing Speed | Advantages                                      | Limitations                                           |
| CNN<br>(Convolutional<br>Neural Network)                  | Deep<br>Learning    | Crop disease<br>detection, image<br>classification | High      | Medium           | Excellent image<br>recognition capability       | Requires large<br>datasets and<br>computational power |
| ResNet                                                    | Deep<br>Learning    | Leaf disease<br>classification                     | Very High | Medium           | Handles deep<br>neural networks<br>efficiently  | Complex<br>architecture                               |
| MobileNet                                                 | Deep<br>Learning    | Drone and mobile-<br>based crop<br>monitoring      | High      | Fast             | Lightweight and<br>suitable for<br>edge devices | Slightly lower<br>accuracy than<br>larger models      |
| YOLO<br>(You Only<br>Look Once)                           | Deep<br>Learning    | Real-time disease<br>and pest detection            | High      | Very Fast        | Real-time object<br>detection                   | Requires extensive<br>training data                   |
| Random Forest                                             | Machine<br>Learning | Crop yield<br>prediction, soil<br>analysis         | Medium    | Fast             | Easy implementation<br>and interpretation       | Less effective for<br>image-based tasks               |
| Support Vector<br>Machine (SVM)                           | Machine<br>Learning | Plant disease<br>classification                    | High      | Medium           | Effective with<br>smaller datasets              | Performance<br>decreases with<br>large datasets       |
| K-Nearest<br>Neighbors (KNN)                              | Machine<br>Learning | Crop<br>classification                             | Medium    | Slow             | Simple and easy<br>to implement                 | Computationally<br>expensive for<br>large datasets    |

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## Performance Comparison

### **CNN (Convolutional Neural Network)**

CNN is the most widely used algorithm for crop disease detection because it can automatically extract image features and identify disease symptoms with high accuracy. It is highly effective in image classification tasks and is commonly used in agricultural AI systems.

### **ResNet**

ResNet improves classification performance by allowing deeper neural network architectures, making it suitable for complex agricultural image datasets. It provides excellent accuracy and is often used in advanced crop disease detection systems.

### **MobileNet**

MobileNet is optimized for deployment on drones, smartphones, and edge devices where computational resources are limited. Its lightweight architecture enables faster processing while maintaining good accuracy.

### **YOLO (You Only Look Once)**

YOLO is ideal for real-time agricultural monitoring because it can detect diseases, pests, and crop anomalies quickly from drone imagery. Its high-speed object detection capability makes it suitable for precision farming applications.

### **Random Forest**

Random Forest is commonly used for prediction tasks such as yield forecasting and soil quality assessment rather than image analysis. It is easy to implement and provides reliable results for agricultural data analysis.

### **Support Vector Machine (SVM)**

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SVM is effective for plant disease classification, particularly when working with smaller datasets. It offers good accuracy but may become less efficient when handling large amounts of data.

**K-Nearest Neighbors (KNN)**

KNN is a simple machine learning algorithm used for crop classification and pattern recognition. While easy to understand and implement, it requires significant computational resources for large datasets.

**Recommended Algorithm for Drone-Based Crop Disease Detection**

| REQUIREMENT                         | RECOMMENDED ALGORITHM |
|-------------------------------------|-----------------------|
| Highest Accuracy                    | ResNet                |
| Real-Time Detection                 | YOLO                  |
| Edge Device Deployment              | MobileNet             |
| General Crop Disease Classification | CNN                   |
| Agricultural Data Prediction        | Random Forest         |

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## Conclusion

AI algorithms have significantly improved agricultural monitoring and crop management. For drone-based crop disease detection, deep learning models such as CNN, ResNet, MobileNet, and YOLO provide superior performance due to their image-processing capabilities. Among these, CNN remains the most widely adopted model, while YOLO is preferred for real-time applications and MobileNet for lightweight deployment on drones and mobile devices. The selection of an algorithm depends on the application's accuracy requirements, processing speed, and available computational resources.